

Cross-Layer Interaction under Weight-Space Ablation: A Closed-Form Attention Jacobian Bound and a Test on a Real Pretrained Model

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Abstract

A companion paper (same author) studies when activation patching and weight-space ablation agree, inside an idealized model in which a conditional computation is carried additively through a residual stream. For the one composition in that model where two carriers are architecturally dependent rather than independent, an attention head and its own layer’s normalization–MLP composition, it derives an exact first-order interaction formula: the interaction is identically zero whenever only the MLP is ablated, and second-order bounded, under a curvature hypothesis, whenever the head is ablated as well. That result is confined to a single residual block. It says nothing about carriers separated by several layers, which is the generic case in any network with more than one block, and everything in it is checked on small transformers trained from scratch on a synthetic conditional task, not on a model built for any other purpose.

This paper takes the single-block result past both limits. First, we show that the interaction produced by ablating an arbitrary subset of carriers spanning several layers decomposes exactly into a sum of same-block terms, one per touched layer, each pinned to zero when that layer’s own carriers are MLP-only and each equal to the companion paper’s bounded interaction term when the layer feeds the readout directly, plus a cross-layer remainder on which the decomposition itself makes no claim of smallness. Second, we isolate that remainder exactly, for two touched layers, as a double integral of a mixed second derivative, the same first-order technique applied one level up, and we name the one ingredient still missing to close it to a usable bound: a Jacobian bound for the attention sub-block itself. We derive that bound in closed form and verify it, pointwise and without a single violation, against Qwen2.5-1.5B-Instruct’s real weights, but we do not yet chain it across many layers, so the identity closes while its numerical scope stays explicitly open. We also supply, in closed form, the curvature constant the companion paper’s own second-order bound leaves unexhibited, computed from the trained weights alone.

Third, on that same model, we search for and find an emergent circuit for indirect object identification, a mechanism nobody trained the model to have and nobody designed into it, using the original activation-patching method for this task. We check the same three-way pattern, collapse, dissociation, and a nonzero interaction, directly on it. The result is genuinely mixed: a carrier shared across all five tested instances does emerge, collapse and dissociation are present on most instances and absent or weak on at least one each, and a nonzero interaction is measurable on three of the five, at layer pairs that sit outside the same-block configuration the companion theorem was proven for, so what they show bears on this paper’s own open multi-layer case rather than on that theorem directly.

1 Introduction

A companion paper (same author) makes exact a mismatch between two families of causal intervention used throughout mechanistic interpretability, activation patching and weight-space

ablation, inside an idealized model of a conditional computation carried additively through a residual stream. Two of its three results, a criterion for when deleting a subset of carriers collapses a matched pair onto a single unconditional branch, and an exact separation between what patching a carrier and ablating it each measure, hold for any residual computation with a linear readout and use no property of transformers. Its third result is different in kind: for the one composition in that model where the independence of carriers is architecturally false, an attention head and its own layer’s normalization–MLP composition, it computes the resulting interaction term exactly, to first order, with a provably second-order remainder.

That third result stops at the boundary of one residual block. A real transformer is a sequential composition of many blocks, and the carriers an experimenter chooses to ablate together are routinely drawn from different layers, sometimes many layers apart. The single-block theorem is silent on this case: it tells us the interaction is exactly zero when an MLP alone is ablated and bounded, at second order, when a head is ablated alongside it, but it says nothing about two heads at different layers, or a head and an MLP several blocks apart. Closing that gap is this paper’s first task, and it matters because the single-block case, while exact, is also the special case least likely to be the one an experimenter actually intervenes on: a component chosen for its causal role is, if anything, more likely to interact with a distant component than with the MLP three lines below it in the same block.

A second gap is empirical rather than mathematical. Every check in the companion paper’s own experiments runs on small transformers trained from scratch specifically to exhibit a clean two-branch conditional. That design choice removes ambiguity about what the ground-truth mechanism is, at the cost of saying nothing about whether the same predictions hold on a mechanism nobody built to have them: an emergent circuit, found rather than designed, inside a model trained on ordinary language for ordinary reasons. Checking the companion paper’s qualitative predictions there, on Qwen2.5-1.5B-Instruct, a genuinely pretrained model at a scale two orders of magnitude past the companion paper’s own instances, is this paper’s second task.

This paper’s three results follow, previewed in the order in which they appear below.

A multi-layer decomposition (Section 4). For an arbitrary ablated subset S spanning any number of layers, the gap between the true, weight-edited network’s selector and the idealized model’s own prediction decomposes *exactly* into a sum of same-block terms, one per touched layer, plus a single remainder on which the decomposition itself makes no claim of smallness. Each same-block term is identically zero whenever that layer’s own ablated carriers are MLP-only, and equals the companion paper’s bounded interaction term exactly whenever that layer feeds the readout directly; the general case, an earlier layer’s discrepancy propagated through everything downstream of it, is isolated precisely enough to be measured on its own, not closed.

An exact cross-layer interaction identity and a closed-form attention Jacobian bound (Section 5). For two touched layers, the remainder above is itself an exact identity: a double integral of a mixed second partial derivative of a two-parameter interpolation between the clean network and the fully edited one, the same fundamental-theorem-of-calculus technique the companion paper uses one level down, applied twice. Closing this identity to a numerical bound reduces to one missing ingredient, since the normalization–MLP half of the relevant Jacobian is already bounded by the companion paper’s curvature machinery: a local operator-norm bound on how a single attention head’s output responds to one perturbed key or value token. We derive that bound exactly and in closed form, and verify it, pointwise on Qwen2.5-1.5B-Instruct’s real weights and a real forward pass, against finite-difference estimates: zero violations across twelve probed sites. What this does not close is composing the bound across every block strictly between the two touched layers; we report the per-layer bound as verified and the chained, multi-layer one as unverified and likely loose, not as closed.

A closed-form curvature constant (Section 6). The companion paper’s second-order interaction bound is conditional on a curvature constant Λ it does not exhibit. We compute Λ in closed form, in terms of the three MLP spectral norms and the normalization scale of the trained weights alone, discharging that hypothesis rather than assuming it.

A real pretrained model’s emergent circuit (Section 7). On Qwen2.5-1.5B-Instruct, an emergent circuit for indirect object identification, found by the original activation-patching method for this task and never designed with this paper’s claims in mind, reproduces the qualitative three-way pattern, collapse, dissociation, and a nonzero interaction, with genuinely mixed fidelity across five tested instances: this section reports every number, including the ones that do not confirm the pattern cleanly.

Section 2 places both extensions relative to existing work. Section 3 restates, compactly and without proof, the pieces of the companion paper’s model and single-block theorem this paper’s own results build on directly. Section 8 separates what this paper establishes from what it leaves open, and Section 9 closes.

2 Related Work

Bounding depth by chaining local Jacobians. Controlling how a local perturbation compounds through the depth of a network by chaining local operator-norm or Lipschitz bounds, one layer at a time, is a standard strategy in the analysis of deep and recurrent networks generally. What this paper needs is a version of that strategy specific to one component that resists it in a well-known way: the dot-product attention sub-block, whose global Lipschitz constant is not bounded in general (the softmax weights depend on the query in a way that a fixed matrix norm does not capture), so that only a *local*, pointwise bound, evaluated at the real activations a real forward pass produces rather than a supremum over all possible inputs, is available at all. We derive exactly such a bound for a single perturbed key or value token in Section 5 below. We are not aware of a specific prior closed-form treatment of this particular quantity, and we would rather say so plainly than attach a citation we are not confident is the right one; the general chaining strategy itself is not new, only this paper’s particular use of it.

Real-model circuit discovery. The method this paper reuses to find an emergent circuit inside Qwen2.5-1.5B-Instruct, a greedy activation-patching search over every attention head and MLP output pruned for redundant sites, is Wang et al.’s original method for locating the indirect-object-identification circuit in GPT-2 small [1]; we apply it unchanged, at a different scale, to a different pretrained model. The pattern we observe on the resulting circuits, a small core shared across lexical instances alongside instance-specific idiosyncratic support, echoes two findings from work on self-repair and redundant representation: the Hydra effect, in which ablating an attention layer causes downstream layers to compensate, measuring an importance score that reflects self-repair as much as the ablated component’s own role [3], and the general expectation, from work on superposition, that a trained network routinely spreads a single computation across more directions and components than the minimum needed to carry it, precisely because doing so buys redundancy [2]. Neither result is about the interaction term this paper isolates specifically; we invoke them only for the shared-plus-idiosyncratic pattern they predict and that this paper’s own circuit search reproduces.

Concurrent work on interaction effects under intervention. A parallel line of work, appearing while this paper was being prepared, reaches the same core observation from the activation-patching side rather than the weight-space side. Vaidyanathan et al. [4] re-derive the activation-patching estimand from causal mediation analysis and show that the natural indirect

effect attributed to a component also contains an interaction term, measuring how much that component’s effect depends on the state of the others; they prove that this term scales with the distance between clean and patched activations, that it is negligible when the model is locally affine, and that it decomposes combinatorially into pairwise and higher-order group contributions. The overlap with the present paper is real and worth stating plainly: both isolate an interaction term that a first-order account of intervention effects silently absorbs. The difference is equally real. Their analysis concerns activation patching and leaves the “locally affine” condition qualitative; the object bounded here is a weight-space ablation, and the closed-form attention Jacobian of Section 5 is precisely a quantitative statement of how far one attention sub-block departs from affine, evaluated pointwise on real weights rather than assumed. In that sense the bound derived here can be read as supplying, for one architecture and one intervention type, the constant their condition leaves implicit. Two further concurrent papers address neighbouring questions empirically rather than in closed form: Gong et al. [5] show that self-repair by dormant backup components corrupts first-order ablation scores, and Guo et al. [6] separate “transports task-relevant content” from “computation degrades when removed” using paired interventions, on GPT-2 and on the same Qwen2.5-1.5B-Instruct model used in Section 7.

What is new here, relative to the companion paper. The companion paper’s own single-block interaction theorem is confined to two carriers inside one residual block and is checked only on synthetic-task transformers. This paper contributes (i) an exact decomposition of the interaction produced by an arbitrarily distributed, multi-layer ablated subset into same-block terms plus one isolated cross-layer remainder (Section 4); (ii) an exact identity for that remainder, for two touched layers, together with the one missing closed-form ingredient needed to bound it, verified pointwise on real weights (Section 5); (iii) the closed-form curvature constant the companion paper’s own second-order bound leaves unexhibited (Section 6); and (iv) a check of the companion paper’s qualitative three-way pattern on an emergent circuit inside a genuinely pretrained model, reported with the same discipline of stating a mixed result as a mixed result rather than rounding it toward confirmation (Section 7).

3 Setup: The Single-Block Model and Result

This section restates, compactly and without proof, the pieces of the companion paper’s abstract conditional model and single-block interaction theorem that this paper’s own propositions build on directly. Nothing here is new. Results the companion paper proves are stated using the **fact** environment and attributed to it in one sentence; notation and definitions are stated as plain prose or in the ordinary **definition/assumption** environments already used for such objects, since restating a definition involves no proof to omit.

Carriers and the residual decomposition. The companion paper’s abstract conditional model posits that a residual mapping decomposes as

$$F(x) = F_0(x) + \sum_{i=1}^k \alpha_i(x) v_i,$$

with F_0 the unconditional part, $v_i \in \mathbb{R}^d$ fixed directions, and $\alpha_i(x)$ scalar selector functions; the pair (v_i, α_i) is the i -th *carrier*.

Assumption 1 (Low-rank support). *The conditional component has finite-dimensional support: a subspace $\mathcal{C} = \text{span}\{v_1, \dots, v_k\}$ with $\dim(\mathcal{C}) \ll d$.*

For an index subset $S \subseteq \{1, \dots, k\}$, deleting the carriers in S replaces F by

$$F_S(x) := F_0(x) + \sum_{i \notin S} \alpha_i(x) v_i, \tag{1}$$

leaving F_0 and every surviving carrier exactly as it was.

Remark 1 (From a weight edit to a deleted term). *The companion paper shows that (1) is exactly what a low-rank edit of one component’s own weights does to the residual stream: an attention head with pre-projection activation $a(x)$ and output projection W writes $Wa(x)$ into the stream, and projecting a unit direction u out of that projection, $W \mapsto W(I - uu^\top)$, changes the write-in by exactly $(Wu)\langle u, a(x) \rangle$, i.e. deletes one term $v\alpha(x)$ with $v := Wu$ and $\alpha(x) := \langle u, a(x) \rangle$. A projector of rank ℓ deletes ℓ such terms. This is an identity, not an approximation, for any single component’s own weight edit; what is not an identity is the further assumption that deleting one component’s term leaves every other component’s term unchanged, which is exactly what Section 5 below revisits across layers.*

The linear readout and matched pairs.

Assumption 2 (Linear readout). *There exist a linear functional $\psi \in (\mathbb{R}^d)^*$ and a bias $b \in \mathbb{R}$ such that the network’s binary decision on $F(x)$ is determined by the sign of $s(x) := \psi(F(x)) + b$.*

Definition 1 (Matched pair). *A matched pair is a pair of inputs (x_A, x_B) for which the network is correct exactly when $g_A := s(x_A) > 0 > s(x_B) =: g_B$; g_A, g_B are the pair’s margins.*

Writing $\beta_i := \psi(v_i)$, ablating S replaces s by the *ablated selector* $s_S(x) := \psi(F_S(x)) + b = s(x) - q_S(x)$, with *removed mass* $q_S(x) := \sum_{i \in S} \beta_i \alpha_i(x)$; for a matched input pair, the companion paper’s collapse theorem gives an exact if-and-only-if criterion, on q_S ’s symmetric and antisymmetric parts across the pair, for when S maps both x_A and x_B to one shared value, and its dissociation theorem gives the following.

Fact 1 (Patching–ablation dissociation). *Patching carrier i from a donor input to a receiver input moves the readout by exactly $\beta_i \delta_i$, with $\delta_i := \alpha_i(x_A) - \alpha_i(x_B)$ the carrier’s contrast; ablating carrier i moves the readout by exactly $-\beta_i \alpha_i(x_B)$, the carrier’s absolute level at the receiver. Neither quantity bounds the other, in general.*

Fact 2 (Single-carrier patch–edit equivalence). *Fix an input x and a single component whose write-in to the residual stream is $Wc(x)$, with $c(x)$ its own activation and W its output matrix. Ablate it by an orthogonal projector, applied either on the activation side (P) or on the stream side (Q). Then the weight-edit route (edit W and run a fresh forward pass) and the frozen-activation route (leave all weights intact, replace this component’s realized output by its projected value, and recompute everything downstream) assign identical values to every node of the network. This holds for any projector, including one estimated from data, since both routes use the same one.*

The two-carrier composition inside one block. The companion paper’s single-block theorem considers two carriers written into the same residual stream by a single block: an attention head with pre-projection activation $a(x)$ and output projection W , contributing $Wa(x)$ to the post-attention residual $r_1(x) := x + \text{ao}(x)$, and the block’s own MLP M , applied after a normalization layer N acting on r_1 . Writing $g := M \circ N$, the second carrier’s realized value $g(r_1(x))$ is a function of the first carrier’s output, not an independent term. Ablating the head with an orthogonal projector $P = UU^\top$ replaces W by $W(I - P)$, equivalently subtracting $\eta(x) := WPa(x)$ from r_1 (Remark 1); ablating the MLP with an orthogonal projector Q on $\mathbb{R}^d$ replaces $M(z)$ by $(I - Q)M(z)$. Let $N(r) := \rho(r)\gamma \odot r$ with $\rho(r) := (\frac{1}{d}\|r\|^2 + \varepsilon)^{-1/2}$ be root-mean-square normalization with scale γ and $\varepsilon > 0$.

Fact 3 (Normalization Jacobian). *N is C^∞ on all of $\mathbb{R}^d$, with*

$$DN(r) = \rho(r) \text{diag}(\gamma) \left(I_d - \frac{\rho(r)^2}{d} r r^\top \right), \quad \|DN(r)\|_{\text{op}} \leq \|\gamma\|_\infty \rho(r).$$

Fact 4 (First-order interaction formula). *Let $\Delta(x) := (I - Q)[g(r_1(x) - \eta(x)) - g(r_1(x))]$ be the gap between the true, jointly-recomputed effect of the head’s and the MLP’s ablation and the frozen-activation estimate that treats the two carriers as independent. For every x ,*

$$\Delta(x) = (I - Q) \left[-Dg(r_1(x)) \eta(x) - R(x) \right],$$

$$R(x) := \int_0^1 \left[Dg(r_1(x) - t\eta(x)) - Dg(r_1(x)) \right] \eta(x) dt,$$

with $Dg = DM(N(r))DN(r)$ by the chain rule; the identity is exact for every x , every $\eta(x)$, and every Q . If $\|D^2g\|_{\text{op}} \leq \Lambda$ on the segment joining $r_1(x) - \eta(x)$ and $r_1(x)$, then $\|R(x)\| \leq \Lambda \|\eta(x)\|^2$: the interaction is second order in the size of the perturbation. In particular, $\Delta(x) \equiv 0$ identically whenever $\eta(x) \equiv 0$ (in particular whenever only the MLP, and not the head, is ablated), and trivially whenever $Q = I$.

Section 6 below supplies Λ in closed form, so this hypothesis is checkable on a trained network’s own weights rather than assumed.

Fact 5 (Propagation to the readout). *Suppose the block above feeds the readout directly, i.e. $F(x) = r_1(x) + g(r_1(x)) + E(x)$ with $E(x)$ collecting $F_0(x)$ and every carrier not drawn from this block. For S consisting only of this block’s own head and/or MLP carriers, the idealized selector $s_S(x)$ and the selector $s_S^{\text{true}}(x)$ of the genuinely weight-edited network agree up to exactly the interaction term of Fact 4:*

$$s_S^{\text{true}}(x) = s_S(x) + \psi(\Delta(x)),$$

$$|\psi(\Delta(x))| \leq \|\psi\|_{\text{op}} \left(\|Dg(r_1(x))\|_{\text{op}} \|\eta(x)\| + L(x) \|\eta(x)\| \right),$$

with $L(x) := \sup_{t \in [0,1]} \|Dg(r_1(x) - t\eta(x)) - Dg(r_1(x))\|_{\text{op}}$. Under Fact 4’s curvature hypothesis,

$$|\psi(\Delta(x))| \leq \|\psi\|_{\text{op}} (\|Dg(r_1(x))\|_{\text{op}} \|\eta(x)\| + \Lambda \|\eta(x)\|^2).$$

These six facts, together with the notation $\eta(x)$, Q , $g = M \circ N$, $r_1(x)$, $s(x)$, $s_S(x)$, and $q_S(x)$ introduced above, are everything this paper’s own propositions need from the companion analysis. Every proof in the sections that follow is new to this paper.

4 A Multi-Layer Decomposition

Fact 5 accounts for the interaction exactly, but only when every ablated carrier is drawn from a single block. We now show that the general case (S spanning several layers) decomposes, exactly, into a sum of instances of Fact 5, one per touched layer, plus a remainder that the two-carrier composition of Section 3 does not, and does not attempt to, cover.

Write the network as a sequence of blocks $l = 1, \dots, n_{\text{layers}}$, block l reading the residual x_l and producing $x_{l+1} = x_l + \text{Block}_l(x_l)$. For an ablated subset S , write $S = \bigcup_l S_l$ with S_l the carriers of S drawn from block l ’s own head and/or its own MLP, and $L(S) := \{l : S_l \neq \emptyset\}$. For $l \in L(S)$, let $F_{S_l}^{(l)}$ denote the network in which *only* block l ’s carriers S_l are weight-edited (every other block, including every other element of S at a different layer, keeps its trained weights), and write $s_{S_l}^{(l)}(x)$ for its selector.

For $l \in L(S)$, define the *same-block discrepancy, propagated to the readout*,

$$\Delta_l^{\text{eff}}(x) := s_{S_l}^{(l)}(x) - s_{S_l}(x),$$

the effect, on the actual final selector, of editing block l ’s carriers S_l in isolation, compared with the idealized prediction for S_l alone; not yet Fact 4’s $\psi(\Delta_l(x))$, since S_l ’s block need not be the one feeding the readout directly.

Proposition 1 (Multi-layer decomposition). *For every matched pair, every $S = \bigcup_l S_l$, and every x ,*

$$s_S^{\text{true}}(x) - s_S(x) = \sum_{l \in L(S)} \Delta_l^{\text{eff}}(x) + R_\times(x), \quad (2)$$

for a remainder $R_\times(x)$ on which (2) makes no claim of smallness. Two facts pin down $\Delta_l^{\text{eff}}(x)$ without a new curvature argument: it is identically zero whenever S_l ablates only block l 's MLP, for every l , regardless of position; and it equals Fact 4's $\psi(\Delta_l(x))$ exactly whenever block l feeds the readout directly, by Fact 5, in particular for the last block of a network with no final normalization, as in Section 7's instances. When S touches only such a block, $|L(S)| = 1$ and $R_\times(x) \equiv 0$.

Proof. Since $q_S(x) := \sum_{i \in S} \beta_i \alpha_i(x)$ (Section 3) sums over the index set S , splitting $S = \bigcup_l S_l$ into disjoint groups splits the sum termwise: $q_S(x) = \sum_l q_{S_l}(x)$ for any such partition, where $q_{S_l}(x) := \sum_{i \in S_l} \beta_i \alpha_i(x)$ is computed, as always, on the clean x ; hence $s_S(x) = s(x) - \sum_l q_{S_l}(x)$, and, applied to S_l alone, $s_{S_l}(x) = s(x) - q_{S_l}(x)$. Define $q_l^{\text{true}}(x) := s(x) - s_{S_l}^{(l)}(x)$, so that by the definition of Δ_l^{eff} above, $q_l^{\text{true}}(x) = s(x) - s_{S_l}(x) - \Delta_l^{\text{eff}}(x) = q_{S_l}(x) - \Delta_l^{\text{eff}}(x)$. Define $s_S^\dagger(x) := s(x) - \sum_{l \in L(S)} q_l^{\text{true}}(x)$: the selector obtained by editing every touched block *in isolation* and combining the results additively, exactly as the idealized model combines independent carriers. Substituting,

$$s_S^\dagger(x) = s(x) - \sum_l (q_{S_l}(x) - \Delta_l^{\text{eff}}(x)) = s_S(x) + \sum_{l \in L(S)} \Delta_l^{\text{eff}}(x).$$

Setting $R_\times(x) := s_S^{\text{true}}(x) - s_S^\dagger(x)$ (a definition, not a further claim) gives $s_S^{\text{true}}(x) - s_S(x) = \sum_l \Delta_l^{\text{eff}}(x) + R_\times(x)$, which is (2); every step so far is regrouping a finite sum and substituting definitions, using no hypothesis on where block l sits. For the first pinning-down fact: if S_l ablates only block l 's MLP, its head's write-in is unedited, so the residual entering block l 's normalization is the same in $F_{S_l}^{(l)}$ as in the clean network; the only change to x_{l+1} is the MLP's own projector acting on that same input, which is exactly what the idealized F_{S_l} also applies to the same clean input, so x_{l+1} agrees between the two, and hence so does every downstream computation and the final selector: $\Delta_l^{\text{eff}}(x) = 0$. For the second: if block l feeds the readout directly, $F_{S_l}^{(l)}$ and F_{S_l} are exactly the objects Fact 5 compares, which gives $s_{S_l}^{(l)}(x) = s_{S_l}(x) + \psi(\Delta_l(x))$, i.e. $\Delta_l^{\text{eff}}(x) = \psi(\Delta_l(x))$. When $|L(S)| = 1$, say $L(S) = \{l\}$, editing block l in isolation *is* editing all of S , so $s_{S_l}^{(l)}(x) = s_S^{\text{true}}(x) = s_S^\dagger(x)$ and $R_\times(x) = 0$ identically. $\square$

Remark 2. *Outside the two pinned-down cases, $\Delta_l^{\text{eff}}(x)$ is exactly the quantity the companion paper's own discussion of cross-layer propagation already flagged as open: the effect, on the final selector, of a same-block interaction at an earlier layer once it has propagated through every subsequent block. This proposition does not close that gap: it isolates the quantity precisely enough to be measured on its own, separately from $R_\times$, which is the contribution we take from it. Section 7 measures both $\sum_l \Delta_l^{\text{eff}}(x)$ and $R_\times(x)$ on configurations where some touched layer is not the last, where neither pinned-down case applies to every term of the sum.*

Remark 3. *$R_\times(x)$ is not a same-block phenomenon under a different name: it vanishes whenever S touches a single layer, and is generically present whenever it touches two or more, regardless of whether any touched layer's S_l pairs a head with its own MLP: two heads at different layers with no MLP anywhere in S already generate a nonzero $R_\times$ in general, since ablating the earlier one changes the residual the later one's own removed mass is computed from, which $F_{S_l}^{(l)}$, by construction, evaluates on the untouched residual instead. Because the architecture is a strictly sequential composition of blocks (block l 's weights play no role in producing $x_1, \dots, x_l$, only in producing x_{l+1} onward), $R_\times(x)$ is entirely a forward effect: an earlier ablated layer can change*

what a later one's carrier reads, but not the reverse, so no separate treatment of two touched layers $l < l'$ versus $l > l'$ is needed beyond tracking, for each touched pair, which one comes first. We do not attempt a closed-form bound on $R_\times(x)$ in this section: doing so requires a curvature bound for the composition of an arbitrary number of further blocks, each contributing its own attention nonlinearity in addition to the normalization-MLP composition Section 6 already bounds, and Section 5 below is exactly the attempt to close that gap, reporting honestly how far it reaches and where it stops. Section 7 measures $\sum_l \Delta_l^{\text{eff}}(x)$ and $R_\times(x)$ separately on the trained instances, using (2) to attribute the measured interaction magnitude of a multi-layer configuration between the two.

5 An Exact Cross-Layer Interaction Identity

Remark 3 states that $R_\times(x)$ is generically nonzero once S touches two or more layers and declines to bound it. This section isolates $R_\times(x)$ exactly, for two touched layers, as the same kind of object $\Delta(x)$ already is in Fact 4 (one level up, across two entire blocks instead of two carriers inside one), and names precisely what stands between this identity and a closed-form bound in the style of Section 6 below.

Fix $S = S_{l_1} \cup S_{l_2}$ touching exactly two blocks, $l_1 < l_2$ (Remark 3: only the order matters). Write $\mu_l(z) := \text{Block}_l(z) - \text{Block}_l^{S_l}(z)$ for the write-in block l 's own weight edit deletes at input z , where $\text{Block}_l^{S_l}$ is block l 's edited map, exactly the block-level analogue of $q_l^{\text{true}}(x) := s(x) - s_{S_l}^{(l)}(x)$ from the proof of Proposition 1, one layer earlier in the computation. Expanding via the identity used in the proof of Fact 5 (there applied to the whole network; here to block l alone) gives the closed form

$$\mu_l(z) = \eta_l(z) + Q_l g_l(r_{1,l}(z)) - \Delta_l(z), \quad (3)$$

with $\eta_l, Q_l, g_l, r_{1,l}$ block l 's own instances of Section 3's notation and Δ_l block l 's own instance of Fact 4's interaction term: the within-block result is not bypassed here, it is exactly what supplies μ_l when block l 's own two carriers are both touched ($\mu_l = \eta_l$ when only the head is, since then $Q_l = 0$ and $\Delta_l = g_l(r_{1,l}(z) - \eta_l(z)) - g_l(r_{1,l}(z))$ cancels the corresponding piece of η_l exactly; $\mu_l = Q_l g_l(r_{1,l}(z))$ when only the MLP is, since then $\eta_l = 0$ forces $\Delta_l \equiv 0$ by Fact 4).

For $(t_1, t_2) \in [0, 1]^2$, let $H(t_1, t_2)$ be the selector of the network in which block l_1 's map is $z \mapsto \text{Block}_{l_1}(z) - t_1 \mu_{l_1}(z)$, block l_2 's is $z \mapsto \text{Block}_{l_2}(z) - t_2 \mu_{l_2}(z)$, and every other block is unedited. H is linear in t_1 and in t_2 separately at each fixed underlying input, and is a composition of C^∞ maps (Fact 3 for every normalization layer; SiLU and softmax are C^∞), hence $H \in C^2([0, 1]^2)$.

Proposition 2 (Exact two-layer cross term). *With H as above, $H(0, 0) = s(x)$, $H(1, 0) = s_{S_{l_1}}^{(l_1)}(x)$, $H(0, 1) = s_{S_{l_2}}^{(l_2)}(x)$, $H(1, 1) = s_S^{\text{true}}(x)$, and*

$$R_\times(x) = H(1, 1) - H(1, 0) - H(0, 1) + H(0, 0) = \int_0^1 \int_0^1 \frac{\partial^2 H}{\partial t_1 \partial t_2}(t_1, t_2) dt_1 dt_2.$$

Proof. The four corner values follow from the definitions of $\text{Block}_l^{S_l}$ and μ_l : $(t_1, t_2) = (0, 0)$ edits nothing, giving $s(x)$; $(1, 0)$ edits only block l_1 's map, which is exactly $F_{S_{l_1}}^{(l_1)}$ by definition of that object; $(0, 1)$ is symmetric; $(1, 1)$ edits both blocks' maps simultaneously and every other block is untouched, which is F_S^{true} . The first equality is then the definition of $R_\times(x)$ used in the proof of Proposition 1, specialized to $|L(S)| = 2$ (there, $s_S^\dagger(x) = s(x) - \sum_l q_l^{\text{true}}(x)$ with two terms, i.e. $-H(1, 0) - H(0, 1) + 2H(0, 0)$; adding $H(1, 1) - H(0, 0)$ and simplifying gives the first equality). For the second: fixing t_1 , the one-dimensional fundamental theorem of calculus gives $H(t_1, 1) - H(t_1, 0) = \int_0^1 \partial_{t_2} H(t_1, t_2) dt_2$; since $H \in C^2([0, 1]^2)$, $t_1 \mapsto \partial_{t_2} H(t_1, t_2)$ is C^1 , and applying the fundamental theorem of calculus again, in t_1 , to $t_1 \mapsto [H(t_1, 1) - H(t_1, 0)]$ gives

$$[H(1, 1) - H(1, 0)] - [H(0, 1) - H(0, 0)] = \int_0^1 \int_0^1 \partial_{t_1} \partial_{t_2} H(t_1, t_2) dt_1 dt_2,$$

and the left side rearranges to $H(1, 1) - H(1, 0) - H(0, 1) + H(0, 0)$. This is exactly the argument used in the proof of Fact 4, applied twice, one dimension up. $\square$

Attention’s own input is likewise normalization-bounded here (ao reads $N(x_l)$, not x_l directly), which raises the question of whether a bound on attention’s own Jacobian, in the point-wise style of Section 6 below (evaluated at the real weights and the real probed input, not a supremum over all possible inputs), is available at all. It is, for a single key/value token perturbed at a time, and we verify it numerically before using it for anything.

Proposition 3 (Local attention Jacobian bound). *Fix a head with query position τ and write $u_i := N(x_i)$ for the normalized input every head reads. Let $q := W_Q u_\tau$, $k_i := W_K u_i$, $v_i := W_V u_i$ for $i \leq \tau$ (causal masking), $p := \text{softmax}(q^\top k_\bullet / \sqrt{d_h})$, and $a := \sum_{i \leq \tau} p_i v_i$ the head’s output at τ . For $j < \tau$,*

$$\frac{\partial a}{\partial u_j} = p_j W_V + p_j (v_j - a) \otimes \frac{W_K^\top q}{\sqrt{d_h}}, \quad (4)$$

exactly, and consequently

$$\left\| \frac{\partial a}{\partial u_j} \right\|_{\text{op}} \leq p_j \|W_V\|_{\text{op}} + p_j \|v_j - a\| \frac{\|W_K\|_{\text{op}} \|q\|}{\sqrt{d_h}} \leq p_j \|W_V\|_{\text{op}} \left(1 + \frac{2d \|\gamma\|_\infty^2 \|W_Q\|_{\text{op}} \|W_K\|_{\text{op}}}{\sqrt{d_h}} \right), \quad (5)$$

the second inequality using $\|u_i\| \leq \sqrt{d} \|\gamma\|_\infty$ (the same bound Section 6’s Proposition 4 uses) to close $\|v_j\|$, $\|q\|$ and $\|v_j - a\| \leq 2 \max_i \|v_i\|$.

Proof. u_j affects a only through $v_j = W_V u_j$ directly and through $k_j = W_K u_j$, which affects p (every logit $z_i := q^\top k_i / \sqrt{d_h}$ for $i \neq j$ is independent of u_j). The direct term contributes $p_j \partial v_j / \partial u_j = p_j W_V$. For the softmax term, the standard softmax Jacobian $\partial p_i / \partial z_j = p_i (\delta_{ij} - p_j)$ and $\partial z_j / \partial u_j = (W_K^\top q)^\top / \sqrt{d_h}$ give $\sum_i v_i \partial p_i / \partial u_j = [p_j v_j - p_j \sum_i p_i v_i] (W_K^\top q)^\top / \sqrt{d_h} = p_j (v_j - a) \otimes (W_K^\top q) / \sqrt{d_h}$, using $\sum_i p_i v_i = a$. Summing the two terms gives (4). The first inequality in (5) is the triangle inequality and submultiplicativity of the operator norm; the second substitutes $\|q\| = \|W_Q u_\tau\| \leq \|W_Q\|_{\text{op}} \sqrt{d} \|\gamma\|_\infty$ and $\|v_j - a\| \leq \|v_j\| + \|a\| \leq 2 \|W_V\|_{\text{op}} \sqrt{d} \|\gamma\|_\infty$ (the last step using $\|a\| \leq \max_i \|v_i\|$, a convex combination of the v_i , and $\|v_j\| \leq \|W_V\|_{\text{op}} \sqrt{d} \|\gamma\|_\infty$ identically). $\square$

Remark 4 (Numerical check). *Both bounds of (5) were checked, in the same refutation-test spirit as Remark 7 below, against finite-difference estimates of $\|\partial a / \partial u_j\|_{\text{op}}$ on Qwen2.5-1.5B-Instruct’s real weights and a real forward pass: 6 (layer, head) pairs spanning layers 1 to 28, 2 key positions each, 24 random unit perturbations per site: 0 violations of either bound in 12 sites. The first (tight) inequality of (5), using the real p_j, v_j, a, q at the probed point, exceeds the empirical estimate by a factor of 12 to 111 (median 71), tighter than Proposition 4’s own looseness (6.5×10^2 to 2.4×10^3) on the same kind of check. The second (weight-only, normalization-closed) inequality is looser by four to six orders of magnitude (median 1.7×10^5 , max 4.2×10^6): finite and never violated, but not a numerically useful estimate as stated. Script: `notebook/verify_attention_jacobian_bound.jl`.*

Remark 5 (What this closes, and what it does not). *Proposition 2 turns “ $R_\times(x)$ is generically nonzero, left open” into an exact double integral of a mixed second derivative, and it isolates precisely one missing ingredient rather than leaving the whole quantity unaddressed. $\partial_{t_1} \partial_{t_2} H$ is nonzero only through how t_1 changes the residual reaching block l_2 ’s input: μ_{l_1} is evaluated at a fixed point, since block l_1 has no edited block upstream of itself, but μ_{l_2} is evaluated at $x_{l_2}(t_1)$, a t_1 -dependent residual once $t_1 > 0$. Bounding $\partial_{t_1} \partial_{t_2} H$ therefore reduces, to leading order, to bounding the operator norm of the Jacobian (with respect to its own input) of the composition of every block strictly between l_1 and l_2 . For each such block, the normalization-MLP half of that Jacobian is already bounded inside the proof of Proposition 4 below (there: $\|DM(z)\|_{\text{op}} \leq$*

$(1 + s_1)\|W_1\|_{\text{op}}\|W_2\|_{\text{op}}\|W_3\|_{\text{op}}\|z\|$ and $\|DN(r)\|_{\text{op}} \leq \|\gamma\|_{\infty}\rho(r)$), and Proposition 3 now supplies the attention half, for a single perturbed key/value token, at the pointwise standard Proposition 4 itself uses, verified rather than merely plausible, per Remark 4. What remains open is composing this one-token, one-layer statement into a bound on the full Jacobian of a composition of an arbitrary number of further blocks: summing (5) over every key/value position gives a per-layer bound (using $\sum_j p_j = 1$, this collapses to $\|W_V\|_{\text{op}}(1+2d\|\gamma\|_{\infty}^2\|W_Q\|_{\text{op}}\|W_K\|_{\text{op}}/\sqrt{d_h})$, independent of the individual p_j), but chaining such per-layer factors across every block strictly between l_1 and l_2 multiplies them together, and whether that product stays controlled (rather than compounding the four-to-six-order-of-magnitude looseness already present in one layer’s closed-form bound into something vacuous after a handful of layers) is exactly the empirical question already raised, in the companion paper’s own discussion of cross-layer propagation, and we do not resolve it here. Unlike the single-layer case, we have not run the corresponding multi-layer refutation test, so we report the per-layer bound as verified and the chained, multi-layer one as unverified and likely loose, not as closed. The same ingredient bears on Gap 2 (the companion paper’s own open question for an intermediate-layer head): H above degenerates to a single-parameter path when $|L(S)| = 1$, and Δ_l^{eff} is ψ applied to exactly the propagated analogue of $R(x)$ in Fact 4: the same per-layer attention bound applies at each step of that propagation, with the same caveat about chaining it across depth.

Remark 6 (No identifiable sign law). Differentiating H twice at $(0,0)$ shows $\partial_{t_1}\partial_{t_2}H(0,0)$ is, to leading order, a bilinear pairing between how much of block l_1 ’s removed mass reaches block l_2 ’s input after propagating through every intervening block, and how sensitively block l_2 ’s own removed-mass map μ_{l_2} depends on its input there, not the product of two scalars this paper has already named. Neither $\Delta_{S_{l_1}} \cdot \Delta_{S_{l_2}}$ nor $\Delta_{S_{l_1}} + \Delta_{S_{l_2}}$ is, in general, the sign of a bilinear pairing between a propagated vector and a Jacobian, so we would not expect either to predict $\text{sign } R_{\times}(x)$ as a rule, a structural reason, not merely an empirical failure to find one. Section 7 gives no support for either candidate: writing $\Delta_H := s(x) - s_H(x)$ and $\Delta_M := s(x) - s_M(x)$ for the two single-site zero-ablations measured there, the identity $-R_{\times}(x) = \Delta_{HM} - \Delta_H - \Delta_M$ (immediate from the definitions, with $S_{l_1} = \{H\}$, $S_{l_2} = \{M\}$) makes Section 7’s already-reported “interaction” column exactly $-R_{\times}(x)$ on real Qwen weights. Its sign matches $\text{sign}(\Delta_H + \Delta_M)$ on two of the three applicable instances (Mary/John, Alice/Bob) and not the third: on Sarah/Tom, $\Delta_H, \Delta_M > 0$ so that candidate rule predicts $-R_{\times} > 0$, but the measured $-R_{\times} = -0.117$. Three points settle nothing statistically, and we report the check only because it was already available at zero additional cost, not as evidence for or against any rule. We were unable to identify a condition, stated in terms of quantities this paper already defines, that predicts $\text{sign } R_{\times}(x)$ in general; the bilinear structure above is a reason to expect that no simple one exists, which we record as a negative finding rather than an unexplored question.

6 A Closed-Form Curvature Constant

Fact 4 assumes a bound $\|D^2g\|_{\text{op}} \leq \Lambda$ on the segment in order to make the remainder second order, but leaves Λ unexhibited. We now compute it, in closed form, from the trained weights. The only ingredient still missing is the second derivative of the normalization; DN already involves $\rho(r)^3$, so D^2N is a third-order tensor, but it collapses to four terms.

Lemma 1 (Second derivative of the normalization). *For every $r \in \mathbb{R}^d$ and all $u, v \in \mathbb{R}^d$,*

$$D^2N(r)[u, v] = -\frac{\rho(r)^3}{d} \gamma \odot (\langle r, v \rangle u + \langle r, u \rangle v + \langle u, v \rangle r) + \frac{3\rho(r)^5}{d^2} \langle r, u \rangle \langle r, v \rangle \gamma \odot r,$$

and consequently $\|D^2N(r)\|_{\text{op}} \leq 6 \|\gamma\|_{\infty} \rho(r)^2 / \sqrt{d}$.

Proof. Differentiating $\partial N_i / \partial r_j = \gamma_i \rho (\delta_{ij} - \frac{\rho^2}{d} r_i r_j)$ from Fact 3 once more, and using $\partial \rho / \partial r_k = -\frac{1}{d} \rho^3 r_k$ throughout, gives

$$\frac{\partial^2 N_i}{\partial r_j \partial r_k} = \gamma_i \left[-\frac{\rho^3}{d} (\delta_{ij} r_k + \delta_{ik} r_j + \delta_{jk} r_i) + \frac{3\rho^5}{d^2} r_i r_j r_k \right],$$

and contracting against $u_j v_k$ yields the displayed identity. For the bound, take $\|u\| = \|v\| = 1$ and use $\|\gamma \odot w\| \leq \|\gamma\|_\infty \|w\|$: the first group is at most $3\|\gamma\|_\infty \rho^3 \|r\|/d$ and the second at most $3\|\gamma\|_\infty \rho^5 \|r\|^3/d^2$. Both are at most $3\|\gamma\|_\infty \rho^2/\sqrt{d}$, since $\rho(r)\|r\| = \|r\|(\frac{1}{d}\|r\|^2 + \varepsilon)^{-1/2} \leq \sqrt{d}$. $\square$

Proposition 4 (Closed-form curvature bound). *Let $M(z) = W_2 h(z)$ with $h(z) := \text{SiLU}(W_1 z) \odot W_3 z$, let $g = M \circ N$, and put $s_1 := \sup_t |\text{SiLU}'(t)|$ and $s_2 := \sup_t |\text{SiLU}''(t)|$. Both are finite universal constants: $s_2 = \frac{1}{2}$, attained at 0, and $s_1 = 1.0998\dots$. Then for every $r \in \mathbb{R}^d$,*

$$\|D^2 g(r)\|_{\text{op}} \leq \|\gamma\|_\infty^2 \rho(r)^2 \|W_1\|_{\text{op}} \|W_2\|_{\text{op}} \|W_3\|_{\text{op}} (6 + 8s_1 + s_2 \sqrt{d} \|\gamma\|_\infty \|W_1\|_{\text{op}}). \quad (6)$$

Consequently Fact 4's curvature hypothesis holds on the segment joining $r_1(x) - \eta(x)$ and $r_1(x)$ with Λ given by the right-hand side of (6) evaluated at $\rho_{\max}(x) := \sup_{t \in [0,1]} \rho(r_1(x) - t\eta(x))$ (a quantity computable from the trained weights and the probed input alone), and, since $\rho \leq \varepsilon^{-1/2}$ everywhere, with a constant uniform in x .

Proof. That $s_2 = \frac{1}{2}$: from $\text{SiLU}(t) - \text{SiLU}(-t) = t\sigma(t) + t\sigma(-t) = t$ we get $\text{SiLU}''(t) = \text{SiLU}''(-t)$, and $\text{SiLU}''(t) = \sigma'(t)(2 + t(1 - 2\sigma(t)))$ is maximized at $t = 0$, where it equals $2\sigma'(0) = \frac{1}{2}$.

Since M is linear in h , $D^2 M(z)[u, v] = W_2 D^2 h(z)[u, v]$ exactly, and writing $a := W_1 z$,

$$D^2 h(z)[u, v] = (\text{SiLU}''(a) \odot W_1 u \odot W_1 v) \odot W_3 z + (\text{SiLU}'(a) \odot W_1 u) \odot W_3 v + (\text{SiLU}'(a) \odot W_1 v) \odot W_3 u.$$

Using $\|p \odot q\| \leq \|p\|_\infty \|q\|$ and $\|p\|_\infty \leq \|p\|$ termwise,

$$\|D^2 M(z)\|_{\text{op}} \leq \|W_2\|_{\text{op}} (s_2 \|W_1\|_{\text{op}}^2 \|W_3\|_{\text{op}} \|z\| + 2s_1 \|W_1\|_{\text{op}} \|W_3\|_{\text{op}}).$$

Similarly $Dh(z)[u] = (\text{SiLU}'(a) \odot W_1 u) \odot W_3 z + \text{SiLU}(a) \odot W_3 u$ and $|\text{SiLU}(t)| \leq |t|$ give $\|DM(z)\|_{\text{op}} \leq (1 + s_1) \|W_1\|_{\text{op}} \|W_2\|_{\text{op}} \|W_3\|_{\text{op}} \|z\|$.

By the chain rule $D^2 g(r)[u, v] = D^2 M(N(r))[DN(r)u, DN(r)v] + DM(N(r)) D^2 N(r)[u, v]$, so

$$\|D^2 g(r)\|_{\text{op}} \leq \|D^2 M(N(r))\|_{\text{op}} \|DN(r)\|_{\text{op}}^2 + \|DM(N(r))\|_{\text{op}} \|D^2 N(r)\|_{\text{op}}.$$

Substituting $\|DN(r)\|_{\text{op}} \leq \|\gamma\|_\infty \rho(r)$ (Fact 3), $\|D^2 N(r)\|_{\text{op}} \leq 6\|\gamma\|_\infty \rho(r)^2/\sqrt{d}$ (Lemma 1), and the uniform activation bound $\|N(r)\| \leq \sqrt{d} \|\gamma\|_\infty$ (which follows from $\rho(r)\|r\| \leq \sqrt{d}$), the first term is at most $\|\gamma\|_\infty^2 \rho^2 \|W_1\| \|W_2\| \|W_3\| (s_2 \sqrt{d} \|\gamma\|_\infty \|W_1\| + 2s_1)$ and the second at most $6(1 + s_1) \|\gamma\|_\infty^2 \rho^2 \|W_1\| \|W_2\| \|W_3\|$. Adding gives (6). $\square$

Proposition 4 has the shape one would hope for: it introduces no architectural quantity beyond the three MLP spectral norms and the γ that already govern the leading-order term through $\|DN\|_{\text{op}} \leq \|\gamma\|_\infty \rho(r)$, and it is multiplied by $\rho(r)^2$ rather than growing with $\|r_1(x)\|$: the MLP's actual input is normalization-bounded, so the remainder cannot blow up merely because the residual stream does.

Remark 7 (Numerical check and looseness). *Both bounds were checked against fourth-point finite-difference estimates of $D^2 N$ and $D^2 g$ on the five trained instances of the companion paper's own marker-task experiments, at real residual-stream points and over random direction pairs: 0 violations in 1200 samples for each. This is a refutation test, not a confirmation: a single violation would indicate an algebraic error above. It is worth being explicit about the price of the product-of-norms argument: on these weights (6) exceeds the largest sampled curvature by a factor of 6.5×10^2 to 2.4×10^3 , while Lemma 1 is loose by only 12.9 – $15.7\times$. The composition step, not the normalization, is where the slack accumulates. Λ is therefore best read as a certificate that the remainder is second order with a computable constant, not as a sharp numerical estimate of it. Script: `notebook/verify_curvature_bound.jl`.*

7 Illustrative Experiments

This section reports two checks. The first reuses the companion paper’s own marker-task checkpoints (five small transformers trained from scratch on a synthetic conditional task, detailed in Appendix A) to measure Proposition 1’s same-block/cross-layer split on real configurations spanning several layers. The second, and the paper’s main empirical result, checks the qualitative three-way pattern of collapse, dissociation, and interaction on a real pretrained model’s own emergent circuit, found rather than designed.

Same-block versus cross-layer. The companion paper’s own experiments fix, per trained instance, three canonical ablation configurations (a single carrier, all carriers in the shallowest carrier-bearing layer, and all carriers) and separately probe every other non-empty subset of each instance’s carrier set, for 39 configurations in total across the five instances; the full interaction-magnitude-versus-idealization-fidelity relationship across all 39 is plotted as a figure in the companion paper, which we do not reproduce here since it is not this paper’s own result. Of those 39, 18 touch two or more layers. Proposition 1 decomposes the interaction magnitude of any subset spanning several layers into a sum of same-block terms Δ_l^{eff} (each identically zero if that layer’s own carriers are MLP-only, and each exactly Fact 5’s already-bounded $\psi(\Delta_l(x))$ only for the layer that feeds the readout directly, an open, propagated quantity otherwise), plus a cross-layer remainder $R_\times$ on which we made no analytical claim either. We measured both sides of (2) on the 18 multi-layer configurations, reusing the same subspaces, and on 80 fresh probe pairs per configuration (1440 pair-configurations in total). The identity itself reproduces to 0 in floating point on every one of them, the same kind of algebraic-identity check used elsewhere in the companion paper’s own robustness analysis. What it lets us attribute is less comfortable than a clean same-block/cross-layer split would suggest. Pooled across all 1440 measurements, the median same-block sum (0.31 logits) and the median total interaction (0.30) are close, but the median cross-layer remainder (0.59) is roughly twice either, and the same-block sum alone exceeds the *total* interaction on 57% of pair-configurations, the cross-layer remainder on 63%, so the two routinely partial-cancel rather than one dominating. The pattern is not uniform: configurations that ablate two heads at different layers with no MLP anywhere in S show the largest same-block sums relative to the total (e.g. a layer-1 and a layer-2 head on instance 2, median same-block 3.85 against a median total of only 0.11; neither layer’s own carriers are MLP-only, so neither term is pinned to zero, and here the two, whatever their individual size, largely cancel against $R_\times$), whereas configurations that pair a head with its own layer’s MLP at one of the touched layers show same-block and cross-layer terms closer in size to the total itself (e.g. seed 22’s layer-1 head+MLP together with a layer-2 head: median same-block 0.21, median total 0.37). The reading we take from this is not that Fact 4 is wrong about the same-block mechanism (the identity that isolates it is exact, and every same-block term we measured nonzero is consistent with that mechanism being active at the layer it sits at) but that for a subset spanning more than one layer, the same-block sum alone is not a reliable proxy for the net interaction: $R_\times$ is routinely the same order of magnitude or larger, and a bound on same-block interaction alone, without one on $R_\times$, would not let a reader anticipate whether the two reinforce or cancel. That bound is the open question Remark 3 states and does not close. Script: `notebook/verify_crosslayer_decomposition.jl`.

7.1 A real pretrained model: an emergent circuit in Qwen2.5-1.5B-Instruct

The companion paper’s own experiments train networks from scratch specifically to exhibit the two-branch structure of Definition 1, and the same-block-versus-cross-layer measurement above reuses those same checkpoints. This final check instead asks whether that structure, and the carriers realizing it, can be found *as-is* inside a real instruction-tuned model that was never trained with either paper in mind: Qwen2.5-1.5B-Instruct (28 layers, 12 query heads, 2 key/value

heads, GQA, $d = 1536$), loaded natively (not through a PyTorch bridge), with logits matching the reference implementation to within 3.7×10^{-5} absolute on the prompts used here.

Finding a matched pair in real language. Definition 1 needs two inputs sharing a fixed selector $s(x) = \text{logit}(c_1) - \text{logit}(c_2)$ with $s(x_A) > 0 > s(x_B)$ and a healthy margin on both, not merely the right sign. Four candidate mechanisms were fixed in advance, before any model was probed, each with 5 lexical instances: indirect object identification (IOI; swapping which of two named entities gave an item to the other, [1]), a sentiment cue (loved/hated $\rightarrow$ positive/negative adjective), temporal order (who left first), and a magnitude comparator (larger/smaller number) kept as a designed-rather-than-found fallback. Only IOI was clean on all 5 instances at margin > 1.0 logit (e.g. Mary/John: $g_A = 8.20$, $g_B = -6.37$); sentiment was clean on 3/5, temporal order on 2/5, and the magnitude comparator on 0/5 (rejected, as flagged in advance). IOI is retained for everything below; the other three families are reported here only because they were pre-registered, not because they matter further.

Method. On the 5 IOI instances, we run a greedy activation-patching search (Wang et al.’s original method for this task, [1]; the same causal-tracing operation this paper uses to reason about carriers throughout) over 364 candidate sites (the output of every attention head (28×12) and every MLP block (28)), using this paper’s own **greedy_patch_search!/backward_prune!** implementation with the recovery metric restricted to $s(x)$ at the final token rather than the full vocabulary, then prunes the found set for redundant sites. Search depth was capped at 6 sites per instance, fixed before any instance was run.

A shared circuit. Figure 1 shows how often each site appears across the 5 pruned circuits. One head, layer 25/head 9, appears in all 5; one more, layer 19/head 6, in 4 of 5; the remaining 16 sites appear in at most 3 circuits each, mostly in exactly one: a small shared core plus instance-specific idiosyncratic support, the same redundant-code pattern reported elsewhere in superposed and self-repairing networks [2, 3] and already discussed in Section 2 above. On every one of the 5 instances, **backward_prune!** removed nothing: all 6 sites found by the greedy search were still individually necessary at its tolerance, so Figure 1 reports circuits with no known-superfluous site left in them. Figure 2 shows the recovery trajectory itself: 4 of 5 instances reach recovery ≥ 0.96 within the 6-site budget; Lucy/Sam plateaus at 0.673 and is the one instance where the search did not find enough of the mechanism within budget.

Three probes, and what operation each one actually performs. On the pruned circuit of each instance we measure three quantities directly analogous to this paper’s central claims, and we state precisely which operation realizes each one before reporting numbers, since the three are not the same kind of intervention.

Collapse. With every site in the pruned circuit forced to 0 (not the frozen-activation prediction used elsewhere in this paper, but a full recomputation of everything downstream, since **patch_node!** invalidates and re-derives every consumer), we compare the selector on x_A and on x_B separately. Forcing a single component’s activation to a fixed value and letting the rest of the network recompute is, node for node, what Fact 2 identifies with editing that component’s own weights by the corresponding projector; here the projector is the identity on the component’s entire output (dimension $d_h = 128$ for a head, so still $\ll d = 1536$ in the sense of Assumption 1; dimension $d = 1536$, the whole residual, for an MLP, so not low-rank in that sense at all, though no result in this paper uses $\dim(C) \ll d$ quantitatively). This is a coarser carrier than the single donor–receiver direction estimated from data in the companion paper’s own experiments, but Remark 1 and Fact 2 are stated for an arbitrary projector, not only a low-rank one, so the single-site case is a literal instance of the deletion (1), not merely an activation-level analogue of it. Jointly forcing several sites to 0 is not separately covered by Fact 2 (stated for

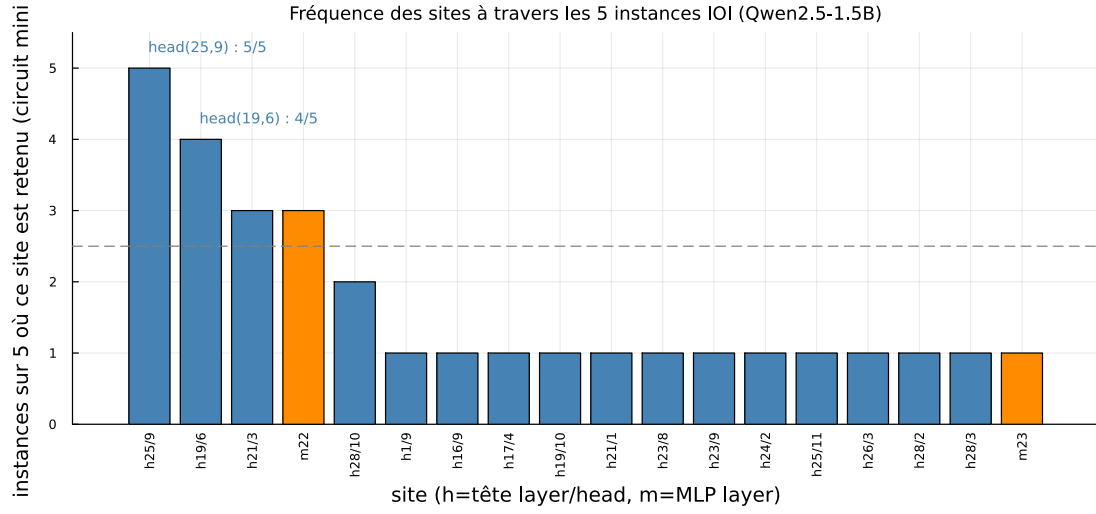


Figure 1: How many of the 5 pruned IOI circuits on Qwen2.5-1.5B-Instruct contain each site (h=attention head layer/head, m=MLP layer). One head (layer 25/head 9) is shared by all 5; one more (layer 19/head 6) by 4; the other 16 sites are idiosyncratic to 1–3 instances.

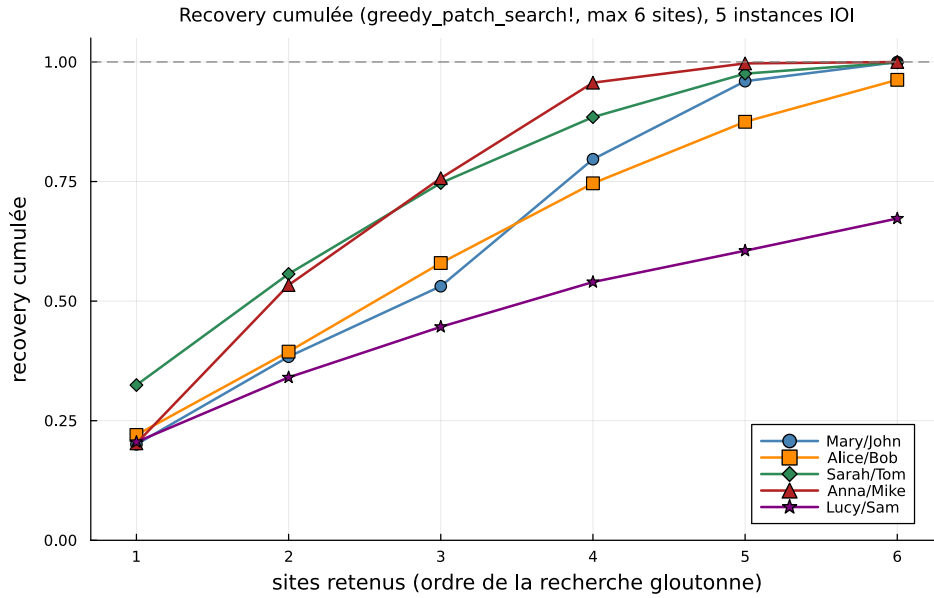


Figure 2: Cumulative recovery of the greedy search, one curve per IOI instance. Four instances converge to recovery ≥ 0.96 by the 6-site budget; Lucy/Sam plateaus at 0.673, the weakest of the five.

one component), though the same argument extends to it: a component whose output is pinned to a fixed value regardless of its own input is, by construction, unaffected by what any other pinned component does upstream of it, so the jointly-recomputed result again coincides node for node with jointly editing every corresponding weight matrix to zero. We did not verify this joint extension independently of that argument, and report it on that basis.

Dissociation. We compare this same joint zero-ablation, measured on x_A ’s own forward pass, against the ordinary activation-patching recovery already used to find the circuit (corrupted run x_B patched toward clean x_A at the same sites). Fact 1 is precisely the claim that patching one input’s activations into another and ablating a component within a single input’s own forward pass are, in general, different operations; here they manifestly are different operations, one substituting a value computed on a different token sequence, the other forcing a component’s own contribution to zero on the same sequence, and we report the gap between their effect sizes as exactly that: a dissociation between two operations, not an approximation error between two estimates of the same one.

Interaction. On the 3 instances whose pruned circuit contains both a head and an MLP, we take the first head and first MLP found (an arbitrary tie-break among the pruned sites, not a choice aimed at any particular layer pair) and measure the true, jointly-recomputed effect of ablating both against the sum of their two individually-measured effects: exactly the frozen-activation-versus-true-network gap this paper calls the *interaction magnitude* above, at the selector level. Two of the three pairs found this way have the MLP carrier in an *earlier* layer than the head carrier (mlp layer 22/23 before head layer 25), so the head’s ablation cannot be reaching that MLP’s own computation at all; the third (Sarah/Tom: head layer 19 before mlp layer 22) has the right order but the two are three transformer blocks apart, not separated by the single RMSNorm of a shared block that Fact 4 formalizes. None of the three pairs instantiates that fact’s same-block head-then-MLP configuration; what nonzero interaction they show instead bears on this paper’s own open cross-multi-layer generalization of Section 4 (Proposition 1, Remark 2), not on the companion paper’s theorem itself. More precisely, writing $\Delta_H := s(x) - s_H(x)$ and $\Delta_M := s(x) - s_M(x)$ for the two single-site zero-ablations and Δ_{HM} for their joint ablation, this “interaction” column is exactly $\Delta_{HM} - \Delta_H - \Delta_M = -R_\times(x)$ of Section 5 with $S_{l_1} = \{H\}$, $S_{l_2} = \{M\}$: these three numbers are not merely analogous to $R_\times(x)$, they are real, non-negligible measured values of it on a genuine 1.5B-parameter pretrained network, at the two specific (non-same-block) layer pairs each instance happens to provide. Remark 6 above uses these same three numbers in its search for a sign rule and finds none.

Instance	Recovery	Collapse ratio	Dissociation gap	Interaction (rel.)
Mary/John	1.000	0.245	0.678	0.124
Alice/Bob	0.963	0.224	0.489	0.327
Sarah/Tom	0.999	0.168	0.546	0.066
Anna/Mike	1.000	0.761	0.852	n/a (0 MLP in circuit)
Lucy/Sam	0.673	0.396	0.224	n/a (0 MLP in circuit)

Table 1: The three probes on the 5 IOI instances. Recovery is `greedy_patch_search!`’s final joint activation-patching recovery (cross-input, no weight-edit reading). Collapse ratio is $|s_{\text{ablated}}(x_A) - s_{\text{ablated}}(x_B)|/|s(x_A) - s(x_B)|$ after jointly zero-ablating the pruned circuit on each input separately (0 = full collapse onto a shared value, 1 = no collapse). Dissociation gap is the absolute difference between the patching-recovery and the same-circuit ablation effect on x_A alone. Interaction (rel.) is $|\Delta_{HM} - \Delta_H - \Delta_M|/(|\Delta_H| + |\Delta_M|)$ for the first head/MLP pair in the pruned circuit, where applicable.

Honest verdict. Collapse is clear-to-partial on 4 of 5 instances (ratio 0.168–0.396) and absent on the fifth: Anna/Mike retains a ratio of 0.761, i.e. ablating its own 6-site circuit barely narrows

the gap between the two branches, the same kind of instance-specific exception the companion paper reports elsewhere on its own marker-task seeds rather than rounds past. Dissociation is never small: the gap between the patching effect and the ablation effect is at least 0.224 and as large as 0.852 on all 5 instances, i.e. these two interventions never coincide on this circuit, consistent with Fact 1’s claim that they are different operations in general, though absent a theorem-derived target value this is evidence of a real, always-present gap rather than a quantitative confirmation. Interaction is measurable on only 3 of 5 instances (the other two circuits contain no MLP at all, so the probe does not apply, not that it was skipped), and among those three it ranges from small (0.066, Sarah/Tom) to non-negligible (0.327, Alice/Bob), and, as argued above, none of the three pairs sits in the same-block configuration Fact 4 was proven for, so this is at most suggestive of this paper’s own open multilayer case, not a test of the companion theorem itself. We report all three probes as a genuinely mixed result on a real model neither trained nor selected for either paper’s claims: a real, cross-instance shared carrier does emerge (Figure 1), collapse and dissociation both occur on most instances and both fail to occur cleanly on at least one, and the interaction probe (the one closest, in its literal operation, to this paper’s own weight-space machinery) lands mostly outside the exact scope any theorem here was proven for.

8 Discussion

What is established. Proposition 1 gives an exact decomposition of the multi-layer interaction into same-block terms plus one remainder, for an arbitrary ablated subset spanning any number of layers, with no approximation anywhere in the identity itself. Two of the three quantities appearing in it are fully pinned down: a same-block term is exactly zero when that layer’s own carriers are MLP-only, and exactly the companion paper’s bounded interaction term when the layer feeds the readout directly. Proposition 2 then isolates the third quantity, the cross-layer remainder $R_{\times}$, exactly, for two touched layers, as a double integral of a mixed second derivative; this is a genuine reduction of an unaddressed quantity to a named, exact object, not a bound. What is established about bounding it is partial by construction: as Remark 3 states, $R_{\times}$ is not itself bounded in this paper, and reducing that question to a chained, multi-block Jacobian bound is as far as we take it. Proposition 3 supplies the one closed-form ingredient this paper identified as missing for that chain, a local attention Jacobian bound verified pointwise, without a single violation, against Qwen2.5-1.5B-Instruct’s real weights; it does not yet chain across many layers, and we report that limitation as open rather than implicitly resolved. Proposition 4 supplies, separately, the closed-form curvature constant the companion paper’s own second-order interaction bound needs and does not itself exhibit, discharging that hypothesis on trained weights rather than leaving it assumed.

What remains open. $R_{\times}(x)$ ’s own closed-form bound is the central open item this paper leaves: Section 5 names the missing ingredient precisely (chaining the per-layer attention and normalization–MLP Jacobian bounds across every block between two touched layers) without closing it, and Remark 6 adds a structural argument, not merely an empirical failure, for why no simple sign rule for $R_{\times}(x)$ should be expected to exist at all. The attention Jacobian bound of Proposition 3 is itself verified only at a single layer and a single perturbed token; whether the per-layer factor compounds into something numerically vacuous after a handful of further blocks, given the four-to-six-order-of-magnitude looseness already present in the weight-only version of the bound at one layer, is an open empirical question this paper does not resolve. Finally, the Qwen2.5-1.5B-Instruct result is a genuinely mixed one on a single pretrained model and a single mechanism (indirect object identification), not the kind of systematic replication across models and tasks that would let us generalize past it: one instance shows no real collapse (Anna/Mike, ratio 0.761), the interaction probe applies to only three of the five circuits, and none of the three

applicable instances sits in the same-block configuration the companion paper’s own theorem was proven for, so what they measure is at most suggestive evidence about this paper’s still-open multi-layer case, not a test of a closed result.

9 Conclusion

A single residual block’s interaction term, exact and provably second-order bounded in the companion paper, says nothing on its own about a network with more than one block. This paper closes part of that gap and states honestly what remains: an exact decomposition attributes the multi-layer interaction to same-block terms, each of them fully accounted for, plus a single cross-layer remainder that is itself isolated exactly rather than bounded. Closing that remainder to a usable numerical bound needed one further ingredient, a Jacobian bound for the attention sub-block, which did not exist in closed form before this paper and which we derive and verify, without a single violation, against a real model’s real weights; chaining that bound across many layers is the piece we did not close, and we say so rather than let the single-layer verification imply more than it shows.

The same real model, probed for a mechanism nobody designed into it, gives this paper’s most direct evidence that the companion paper’s qualitative predictions are not an artifact of training a small transformer to have them. That evidence is mixed by construction: a shared carrier across instances, collapse and dissociation present on most of five tested instances and genuinely absent on at least one, and a nonzero interaction measurable on three of them, at layer pairs the companion theorem was never proven to cover. We take the mixed result as the more informative one. A theory that only ever gets confirmed on the networks built to confirm it is not yet a theory of anything else, and the honest next question this raises, whether the same pattern holds across further models, further mechanisms, and enough instances to test the sign question we could not settle here, is sharper for having been asked on a model that was never asked to answer it.

A Experimental Details and Reproducibility

This appendix gives what is needed to reproduce the same-block-versus- cross-layer measurement of Section 7, which reuses the companion paper’s own marker-task checkpoints; the Qwen2.5-1.5B-Instruct subsection of Section 7 is otherwise self-describing.

Task. Key and value tokens are drawn from a vocabulary of $V = 8$ symbols, with two additional marker tokens m_A, m_B , so the model’s vocabulary has 10 entries. Each sequence presents 3 key–value pairs in random order (6 tokens) followed by a marker and a displayed token, giving sequence length 8; the target is the value of the true key, read at the final position.

Architecture and training. Each instance is a 4-layer decoder transformer in the Llama style: model width $d = 64$, 4 attention heads (head dimension 16), SwiGLU MLP with hidden width 128, pre-normalization by RMSNorm with $\varepsilon = 10^{-6}$, learned positional embeddings, weights in float32. There is no final normalization: the unembedding is applied directly to the residual stream, so the logit contrast is an exactly affine functional of $F(x)$ and Assumption 2 holds exactly, rather than approximately, for these networks. Four instances use initialization seed $S \in \{11, 22, 33, 44\}$; the fifth (“inst. 2”) was trained earlier under the same configuration with a different seed. No model was retrained for the measurement reported here: it reads the same five checkpoints the companion paper uses for its own marker-task experiments.

Carrier selection and ablation subspaces. The candidate set is every attention head and every MLP output, $4 \times (4 + 1) = 20$ sites per instance. A site is a carrier when its mean donor-to-receiver patch recovery $r \geq 0.25$ over 8 matched clean pairs. The companion paper’s three canonical configurations per instance are D1 (the highest- r carrier in the shallowest carrier-bearing layer, alone), DJ (all carriers in that layer), and DJA (all carriers); together with every other non-empty subset of each instance’s carrier set, this gives 39 configurations across the five instances, of which 18 touch two or more layers and are the ones this paper’s own measurement uses. For each carrier, ablation subspaces are the leading left singular vectors of donor–receiver activation differences collected over 32 matched pairs, taking the smallest k reaching 90% of the squared-singular-value energy, capped at $k \leq 4$ for a head and $k \leq 8$ for an MLP. The weight edit is $W_O[:, \mathcal{H}] \leftarrow W_O[:, \mathcal{H}](I - UU^\top)$ for a head with column block $\mathcal{H}$, and $W_2 \leftarrow (I - UU^\top)W_2$ for an MLP: these are the concrete instances of (1) referred to in Remark 1, with the directions estimated from data.

Probe seeds. The same-block/cross-layer decomposition uses probe seeds 20260803/20260804 (80 matched pairs per configuration, restricted to the 18 configurations spanning two or more layers), disjoint from every other seed used anywhere in the companion paper.

Code and data availability. All experiments run against the research repository accompanying this submission; no external data is used, and Qwen2.5-1.5B-Instruct’s weights are loaded locally rather than downloaded during any run reported here. The relevant files are:

`notebook/verify_crosslayer_decomposition.jl`

Proposition 1, the same-block/cross-layer split on the 18 multi-layer marker-task configurations.

`notebook/verify_curvature_bound.jl`

Proposition 4, the finite-difference refutation test of Remark 7.

`notebook/verify_attention_jacobian_bound.jl`

Proposition 3, the finite-difference refutation test of Remark 4 on Qwen2.5-1.5B-Instruct’s real weights.

`notebook/code4_real_llm_step1_matched_pair_search.jl`

the search over the four candidate mechanisms and their lexical instances that identifies IOI as the clean matched-pair family on Qwen2.5-1.5B-Instruct.

`notebook/code4_step2_ioi_patching.jl`

the greedy activation-patching circuit search (`greedy_patch_search!/backward_prune!`) and the three probes (collapse, dissociation, interaction) of Table 1.

`notebook/generate_code4_step2_figures.jl`

Figures 1 and 2, generated directly from the JSON results of the preceding script.

The companion paper’s own scripts train and select the carriers for the five marker-task checkpoints this paper’s same-block/cross-layer measurement reuses. These checkpoints are stored under `notebook/marker_ckpt/`:

`notebook/marker_task_experiment.jl`

task, architecture, and training.

`notebook/marker_seed_matrix.jl`

per-seed training and carrier sweep.

Each script above writes a JSON results file that the corresponding figure script reads directly, so no number in a figure is transcribed by hand.

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